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Graph Digital Twin Project Guide

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Contents

Skipping the physical hardware-in-the-loop (HIL) setup is a highly strategic decision given your tight timeline. Physical co-simulation setups involving real-time simulators (like OPAL-RT or Typhoon HIL) and external microcontrollers (like Raspberry Pi) require months of testing just to resolve communication latencies, protocol offsets, and packet-drop synchronization issues. Transitioning to a **purely software-based, data-driven simulation approach** will allow you to construct and validate your machine learning model rapidly, paving the way for a strong, timely IEEE submission.

To address your advisor's feedback directly, we must:

- **Decouple the domains:** Focus strictly on the **Smart Grid** environment (dropping the Industry 5.0 paradigm) to preserve physical consistency under AC power laws.
- **Narrow the scope:** Refine "cyber resilience and active mitigation" down to **Stealthy False Data Injection Attack (FDIA) Detection and Localization**.
- **Define the scale:** Model a **substation or system-level** electrical grid.

Here are the two best implementation options you can pursue right now, complete with their base papers, datasets, and step-by-step plans.

Option A: The Spatiotemporal Graph Neural Network (GNN) Path

This option is highly recommended if you want to keep the "Graph Digital Twin" focus of your original proposal, which is extremely attractive to high-impact journals like IEEE Transactions on Smart Grid.

- **Base Paper:** *"Topology-Aware Cyberattack Detection in Smart Grids Using Explainable Graph Neural Networks".*
- **Primary Dataset:** Synthetic AC power flow data generated via the open-source **pandapower** Python library on the standard **IEEE 14-bus test system**.
- **Required Tools:** Python, `pandapower` (for grid modeling), `PyTorch Geometric` (for GNN models), and `GNNExplainer` (for XAI).

Step-by-Step Implementation Plan:

- **Grid Representation:** Model the IEEE 14-bus system as an undirected graph $G = (V, E, W)$, where V represents the substation buses (14 nodes), E represents transmission branches, and $W \in \mathbb{R}^{14\times14}$ is the weighted adjacency matrix derived from the grid's admittance matrix.
- **Benign Data Generation:** Use `pandapower` to run non-linear AC power flow simulations using historical load profiles (such as public NYISO load data). This generates node features for each bus, representing voltage magnitudes, phase angles, active power, and reactive power:

$$x_i = [V_i, \theta_i, P_i, Q_i]^T$$

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- **Attack Injection (FDIA):** Programmatically inject stealthy FDI attacks. A stealthy attack vector a is designed using the Jacobian matrix of the state estimation equations:

$$a = h(x + c) - h(x)$$

. This alters physical readings but ensures the residual remains below bad data detection (BDD) thresholds, making it "unobservable" to classical estimators.

- **Model Architecture:** Build a 2-layer GraphSAGE or GATv2 network using `PyTorch Geometric`. The model will aggregate spatial correlations from neighboring buses to output a graph-level classification (benign vs. attacked).

- **Interpretability Layer:** Integrate `GNNExplainer`. When an attack is flagged, `GNNExplainer` will identify the compact subgraph and specific node features (e.g., active power injection at Bus 9 that contributed most to the anomaly prediction, providing clear visual evidence for grid operators.

Option B: The Heterogeneous System Logs Path

This is the fastest possible option. Choose this if your time is extremely limited (e.g., less than a month) and you want to work with a pre-existing, fully labeled, real-world benchmark dataset without writing power flow simulations from scratch.

- **Base Paper:** *"Explainable AI for Smart Grid False Data Injection Attack Prediction".*

- **Primary Dataset: Mississippi State University (MSU) and Oak Ridge National Laboratory (ORNL) Power System Attack Dataset**. *(Highly accessible, multi-class CSV files representing 37 power system scenarios, including short-circuits, line maintenance, and cyberattacks like remote command injection and data spoofing).*

- **Required Tools:** Python, `scikit-learn`, `xgboost`, and `SHAP`.

Step-by-Step Implementation Plan:

- **Data Acquisition:** Download the multiclass or binary CSV files from the MSU/ORNL Power System repository.

- **Feature Preprocessing:** Prepare the heterogeneous dataset consisting of 128 features. These features combine high-speed PMU measurements (voltage/current phasor magnitudes and angles) from 4 substations alongside 12 categorical security logs, Snort rules, and relay trip events.

- **Feature Selection:** Apply a tree-based feature selection method to reduce dimensionality and extract the most informative indicators, minimizing computational overhead.

- **Model Training:** Train an ensemble classifier (such as **Extra Trees** or **XGBoost**) to classify snapshots into three classes: Natural Disturbance, No Event, or Malicious Attack.

- **Local Explainability (XAI):** Deploy `TreeSHAP` to calculate game-theoretic feature attribution scores. This will break open the "black box" by generating force plots mapping exactly which substation relays or specific current parameter manipulations triggered the attack alarm.

Strategic Recommendation for Your Meeting on Thursday

When you present to your advisor on Thursday, you can confidently steer the project forward by stating:

"To address your feedback on narrowing the scope and decoupling the domains, I have decided to focus **strictly on the Smart Grid domain** to ensure physical AC modeling consistency. Furthermore, to deliver a high-impact model within our tight timeline, I am skipping the complex physical/hardware laboratory setup and pursuing a **100% data-driven, software-based digital twin framework**. I am focusing on substation-level state tracking, specifically detecting stealthy False Data Injection Attacks."

- If your advisor wants a **highly innovative, topological approach** matching your original slides, propose **Option A** using `pandapower` and PyTorch Geometric, citing the *IEEE Transactions on Smart Grid* benchmark papers.
- If your advisor prioritizes a **fast, guaranteed implementation** with real-world cyber-physical datasets, propose **Option B** using the MSU/ORNL dataset and SHAP explainability.